

What's New: Python Scripting for Adroit Smart UI

Adroit 11 Smart UI | Release highlight

Unlock the Power of Python in Adroit Smart UI

Adroit Smart UI now includes native Python scripting support, enabling engineers and developers to automate tasks, extend system functionality, and build sophisticated solutions using one of the world's most popular programming languages.

Python scripting is available throughout the Smart UI platform, including Graphic Form scripting, Script Spiders, and the Script Datasource, allowing users to implement custom logic wherever it is needed.

Python scripts can run directly within Adroit Smart UI while interacting seamlessly with datasources, dataelements, graphic forms, vectors & controls, spiders and application services.

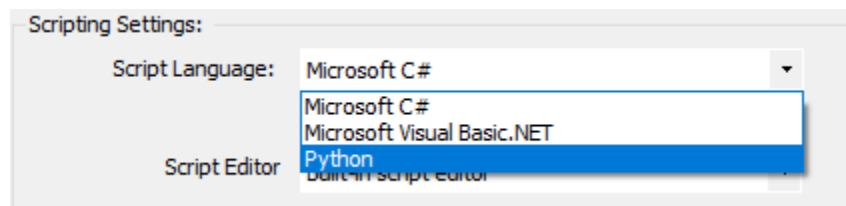


Figure 1: Selecting Python as your Script Language from your profile

Key Benefits

Accelerate Development

Leverage Python's simplicity and extensive ecosystem to create automation, calculations, integrations, and custom business logic faster than ever before.

Access Smart UI Objects Directly

Python scripts have access to Smart UI datasources, dataelements, graphic forms, vectors & controls, spiders, application settings, communications interfaces, and logging services, allowing developers to build powerful solutions without leaving the platform.

Use Your Preferred Development Environment

Developers can edit scripts directly in Smart UI with the built-in Script Editor or launch them into Microsoft Visual Studio or Visual Studio Code for a richer editing and debugging experience, including IntelliSense, breakpoints, and variable inspection.

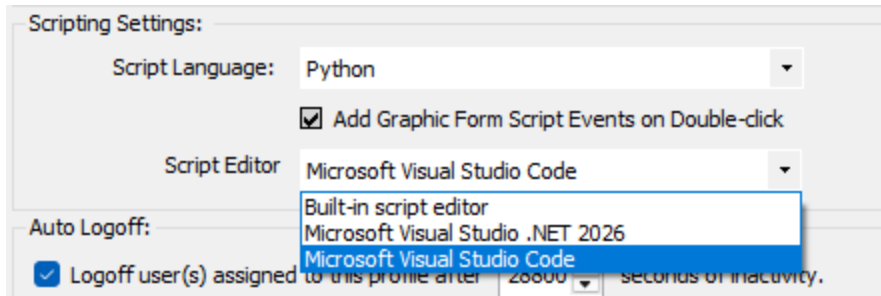


Figure 2: Selecting Visual Studio Code as your external script editor from your profile

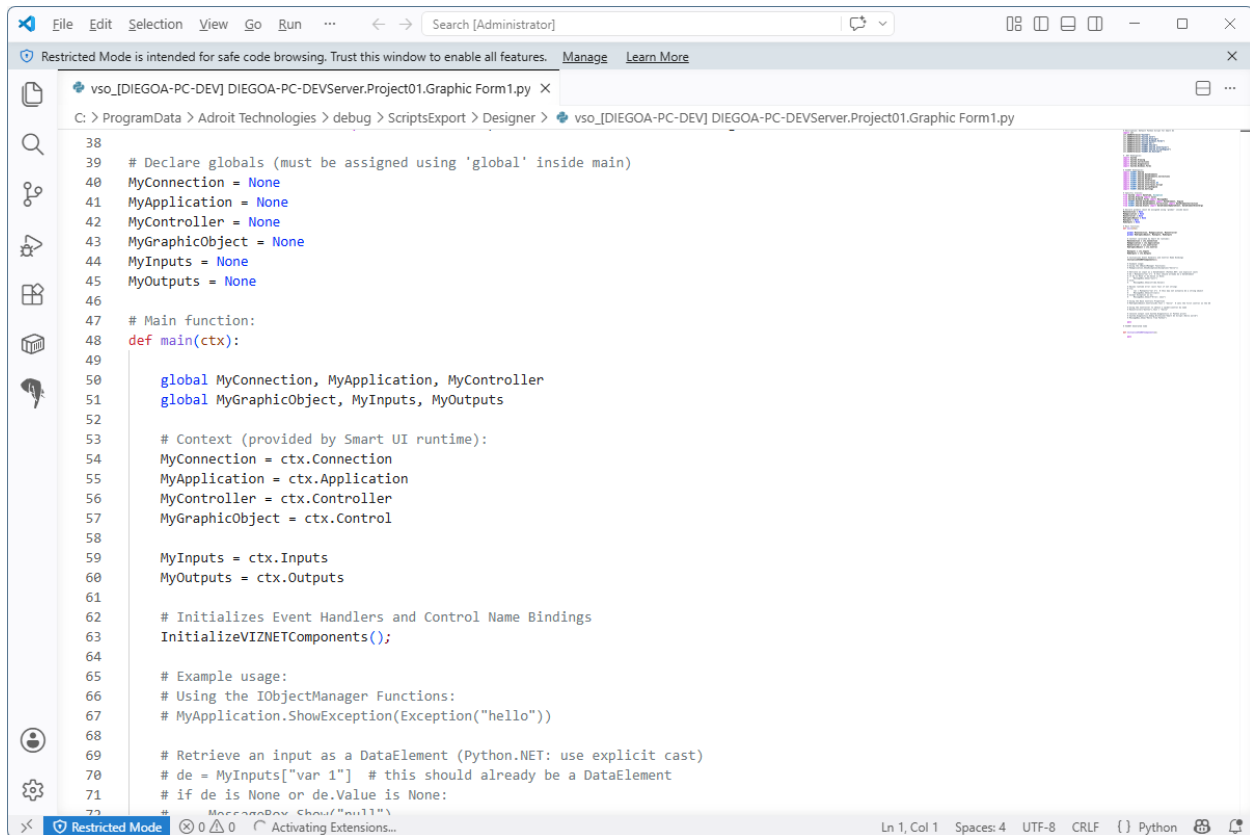


Figure 3: Graphic Form script opened in Visual Studio Code

Built-in Python Package Manager

Extend your scripting capabilities without leaving the application. The built-in Python Package Manager allows users to easily install, update, and remove Python packages directly from the built-in Script Editor, simplifying the setup of third-party libraries and reducing development time.

Key Benefits

- Install and manage Python packages from within the application.

- Quickly add functionality using popular open-source libraries.
- Eliminate the need for external package management tools.
- Streamline script development and deployment workflows.
- Maintain control over installed dependencies in a centralized location.

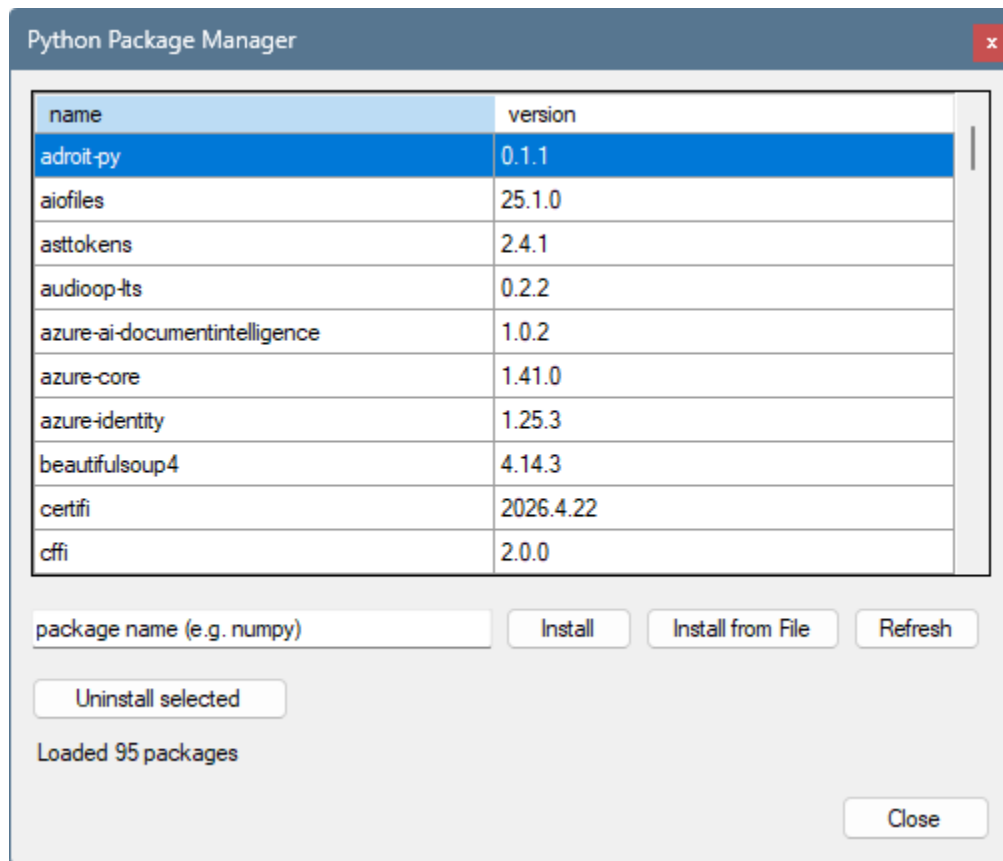


Figure 4: Built-in Python Package Manager launched from the Built-in Script Editor

Simplified Script Management

Automatic file monitoring and script reloading ensure that updates are detected and applied quickly, reducing downtime during development and maintenance.

Enterprise-Ready Reliability

Scripts execute within a controlled environment with error reporting, logging, timeout controls, and safeguards designed to protect the stability of the host application.

Typical Use Cases

- Advanced calculations and data transformations
- Automated workflows and business logic
- Alarm handling and event-driven actions
- Database integration and report generation
- REST API and web service integration
- Custom operator dashboards and user interactions
- Scheduled maintenance and housekeeping tasks
- Historical data analysis and KPI reporting
- Integration with MES, ERP, CMMS, and other enterprise systems
- Predictive analytics and condition monitoring

These capabilities are further enhanced through support for popular Python libraries and frameworks.

Developer Productivity Features

Python scripting includes:

- Built-in Python editor support
- Syntax highlighting and code folding
- Integration with Visual Studio and Visual Studio Code
- IntelliSense support
- Built-in Python Package Manager
- Breakpoint debugging (with the help of python packages or Visual Studio extensions)
- Runtime error reporting
- Automatic script reload on file changes
- Server-side script scheduling and execution management

Why It Matters

Python has become the preferred language for automation, data science, artificial intelligence, systems integration, and rapid application development. By bringing Python directly into Adroit Smart UI, organizations can leverage existing Python skills and libraries while maintaining full access to their operational technology environment.

The result is faster solution delivery, greater flexibility, and a significantly expanded integration ecosystem for Smart UI applications.